

**LECTURE / SESSION<sup>1</sup> PLAN**

| <b>Name Lecturer:</b><br>Sofia Gil-Clavel                                   | <b>Date session:</b><br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Expected number of students:</b><br>20 |
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| Course title                                                                | R tools for teaching and research                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |
| Topic of lecture or session                                                 | Version-controlled repos for courses and research.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                           |
| Situational factors (e.g. group size, prior knowledge, expected motivation) | <p>The class lasts 3hrs.</p> <p>The students have followed all the previous sessions of this course.</p> <p>The students use different Operating Systems (Windows, Mac, or Linux).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                           |
| Intended learning outcomes of this session                                  | <p>At the end of the session the students will be able to:</p> <ol style="list-style-type: none"> <li>1. Explain what Git is.</li> <li>2. Explain what GitHub and CodeBerg are.</li> <li>3. Able to create repository in GitHub or CodeBerg.</li> <li>4. Able to use R-markdown to publish teaching or research material in GitHub or CodeBerg.</li> <li>5. Able to perform the basics of version control: push, pull, and retrack.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                      |                                           |
| Learning material (book, chapters, ...)                                     | <p>The lecture is based on these sources:</p> <ul style="list-style-type: none"> <li>• Chacon, Scott, and Ben Straub. <i>ProGit</i>. 2nd ed. Apress, 2026. <a href="https://git-scm.com/book/en/v2">https://git-scm.com/book/en/v2</a>.</li> <li>• “Getting Started with Codeberg   Codeberg Documentation.” Accessed July 21, 2026. <a href="https://docs.codeberg.org/getting-started/">https://docs.codeberg.org/getting-started/</a>.</li> <li>• Davidson, Polly. “GitHub for Beginners: Your Roadmap to Mastering the GitHub Essentials.” <i>The GitHub Blog</i>, July 15, 2026. <a href="https://github.blog/developer-skills/github/github-for-beginners-your-roadmap-to-mastering-the-github-essentials/">https://github.blog/developer-skills/github/github-for-beginners-your-roadmap-to-mastering-the-github-essentials/</a>.</li> </ul> |                                           |

<sup>1</sup> A session means a teaching and learning session for university students in the bachelor or master. It can be a lecture, a seminar or a specific type of educational meeting with a group of students that is relevant for the discipline where the lecturer can demonstrate a whole range of teaching skills. It means that in one session each of the six didactic elements appears at least once. Your lesson plan should show that you are using a powerful learning environment and that students are activated.

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|                          | <ul style="list-style-type: none"> <li>GitHub Docs. "Creating a GitHub Pages Site." Accessed July 21, 2026. <a href="https://docs-internal.github.com/en/pages/getting-started-with-github-pages/creating-a-github-pages-site">https://docs-internal.github.com/en/pages/getting-started-with-github-pages/creating-a-github-pages-site</a>.</li> </ul> |
| Media, equipment, tools  | <p>The students use their own laptops.</p> <p>The teacher needs access to a projector and a whiteboard.</p>                                                                                                                                                                                                                                             |
| Preparation for students | <p>The students have access to the slides and codes before the class:</p> <p><a href="https://github.com/SofiaG11/Course_Rtools/4_PublishRepo">https://github.com/SofiaG11/Course_Rtools/4_PublishRepo</a></p>                                                                                                                                          |

| Time <sup>2</sup><br>(min.) | Didactic element<br>(goal) <sup>3</sup> and topic <sup>4</sup>                                            | What the teacher does <sup>5</sup><br>(teacher activity)                                                                                     | What students do<br>(learner activity)                                                                          | Evaluation <sup>6</sup><br>(feedback/assessment)                                                                                                                                                                                                 |
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| 10                          | <ul style="list-style-type: none"> <li>Recap of the previous sessions: R-Markdown and servers.</li> </ul> | <ul style="list-style-type: none"> <li>The teacher uses the slides to remind the students what they have learned.</li> </ul>                 | <ul style="list-style-type: none"> <li>The students passively digest what the teacher is explaining.</li> </ul> | <ul style="list-style-type: none"> <li>The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.</li> </ul> |
| 10                          | <ul style="list-style-type: none"> <li>Understand and explain what Git is.</li> </ul>                     | <ul style="list-style-type: none"> <li>The teacher uses the slides to explain what Git is and how it facilitates version control.</li> </ul> | <ul style="list-style-type: none"> <li>The students passively digest what the teacher is explaining.</li> </ul> | <ul style="list-style-type: none"> <li>The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these</li> </ul>                                                             |

<sup>2</sup> Indicate the planned duration in minutes

<sup>3</sup> State the number(s) of the relevant ILO at each didactic element

<sup>4</sup> Only key words

<sup>5</sup> Specify the type of activity and write down also the questions that you prepared to ask in the session

<sup>6</sup> Specify the type of evaluation, i.e. the way in which you assess if the objective(s) has/have been achieved

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|    |                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                            | confusions the teacher will be able to correct the student.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 10 | <ul style="list-style-type: none"> <li>Understand and explain what GitHub and CodeBerg are.</li> </ul>         | <ul style="list-style-type: none"> <li>The teacher uses the slides to explain what GitHub and CodeBerg are.</li> </ul>                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>The students passively digest what the teacher is explaining.</li> <li>The students follow the steps in their computers.</li> </ul> | <ul style="list-style-type: none"> <li>The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.</li> <li>The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.</li> </ul> |
| 40 | <ul style="list-style-type: none"> <li>Able to create a repository either in GitHub or in CodeBerg.</li> </ul> | <ul style="list-style-type: none"> <li>The teacher explains how to create an account in each of them.</li> <li>The teacher explains what a repository is and how to create one.</li> <li>The teacher explains the different elements of a repository while showing the students how to create them using the graphic user interface: <ul style="list-style-type: none"> <li>README</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>The students passively digest what the teacher is explaining.</li> <li>The students follow the steps in their computers.</li> </ul> | <ul style="list-style-type: none"> <li>The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.</li> <li>The teacher walks around the classroom to check on the students and provides feedback when something is</li> </ul>                                                                                                                                   |

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|    |                                                                                                                              | <ul style="list-style-type: none"> <li>○ gitignore</li> <li>○ license</li> </ul>                                                                                                                                                                                                                     |                                                                                                                                                                                | not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.                                                                                                                                                                                                                                                                                                                                                                       |
| 30 | <ul style="list-style-type: none"> <li>• Able to interact with their GitHub or CodeBerg repository using RStudio.</li> </ul> | <ul style="list-style-type: none"> <li>• The teacher explains how to access their GitHub or CodeBerg accounts from RStudio.</li> <li>• The teacher explains how to clone a repository.</li> </ul>                                                                                                    | <ul style="list-style-type: none"> <li>• The students follow the steps in their computers.</li> </ul>                                                                          | <ul style="list-style-type: none"> <li>• The teacher walks around the classroom to check on the students and provides feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.</li> </ul>                                                                                                                                                                                                           |
| 10 | <ul style="list-style-type: none"> <li>• Able to use R-Markdown to create markdown files for their repository.</li> </ul>    | <ul style="list-style-type: none"> <li>• The teacher uses the slides to explain how to use R-Markdown to create a markdown file.</li> <li>• The teacher explains how to format the file for internet users.</li> <li>• The teacher explains how to upload the file to GitHub or CodeBerg.</li> </ul> | <ul style="list-style-type: none"> <li>• The students passively digest what the teacher is explaining.</li> <li>• The students follow the steps in their computers.</li> </ul> | <ul style="list-style-type: none"> <li>• The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.</li> <li>• The teacher walks around the classroom to check on the students and provides feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.</li> </ul> |

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| 40 | <ul style="list-style-type: none"><li>• Able to perform the basics of version control: push, pull, and retrack.</li></ul> | <ul style="list-style-type: none"><li>• The teacher explains what version control is and the different vocabulary around it.</li><li>• The teacher explains how to push, pull, and retrack changes in a repository.</li></ul> | <ul style="list-style-type: none"><li>• The students passively digest what the teacher is explaining.</li><li>• The students follow the steps in their computers.</li></ul> | <ul style="list-style-type: none"><li>• The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.</li><li>• The teacher walks around the classroom to check on the students and provides feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.</li></ul> |
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